

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHM 427 — Past Questions & Materials
by Renance

Course: CHM 427

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 4

Levels: 400

This study pack was compiled and restructured by Renance for CHM 427 students at Federal University of Petroleum Resources, Effurun. It brings together 4 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app

Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 4 — CHM 427 - Process Chemistry

Level 400 • Second Semester • PastQuestion

DEPARTMENT OF CHEMISTRY

COLLEGE OF SCIENCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES

CES

SECOND SEMESTER EXAMINATIONS

CHM 427: PROCESS CHEMISTRY

INSTRUCTIONS: ANSWER ANY FOUR QUESTIONS

UNIT:2 CREDITS TIME:2HRS

1. a. Find the limiting reagent and reactant in excess :

; when 45.421, of CO

with 11.36L of O₂ (g) at STP (0°C and 100kPa) is

1000kg of 8% by weight of NaOH solution ; i

: solution is required, y ion i

water is available. How much of each is required? See also to

2. a. 2.25g of aluminum sulphate reacts with 3.25g of sodium phosphate to make 1,

a precipitate, What is the percent yield? a a ie

b. Three raw materials are mixed in

respectively. The first raw material contains C while the third contains

continuous one at steady state, find the

- a. In the carbonation of coke, the total quantity of carbon dioxide required is the equivalent of 2.5 volumes of gas to one volume of water at 0°C and atmospheric pressure. Calculate

(i) the mass fraction and (ii) the mole fraction of the CO₂ in the drink, ignoring all components other than CO₂ and water

(b) How many moles of C₂H₅OH are there in 50kg of C₂H₅OH

4. a. When fat is removed from whole milk, skim milk is produced. This skim milk comprises 85% water, 65% protein, 7% carbohydrate, 0.5% fat and 1% ash. If the original milk contains 3.5% fat, calculate its composition assuming it was only fat that was removed to make the skim milk and there were no losses

6. 200 moles of SO₂ and 100 moles O₂ are fed to a reactor. Only 100 moles of SO₂ react according to 2SO₂ (g) + O₂ (g) = 2SO₃ (g), find fractional conversion, percentage conversion and fraction unreacted

5. 1000kg/hr of a mixture of benzene (B) and toluene (T) that contained 50% by mass are separated by distillation into two fractions. The mass flow rate of benzene in the product top stream is 450 B/hr and that of toluene in the bottom stream is 475kgT/hr. Assuming a

steady state operation (a) draw and label appropriately a flow chart of the process Gi) write balances on benzene and toluene to calculate the unknown component flow rate in the output streams and give the final flow chart of the process

gb. How can you prepare 1L of 70% of methylated spirit that solely contains isopropano from a 100% isopropy! alcohol.

a tank to give a final product in the ratio 1:0.5:1.5

ntains A and B with 40% A. The second raw material

A and C with 80% A. assuming the process is a

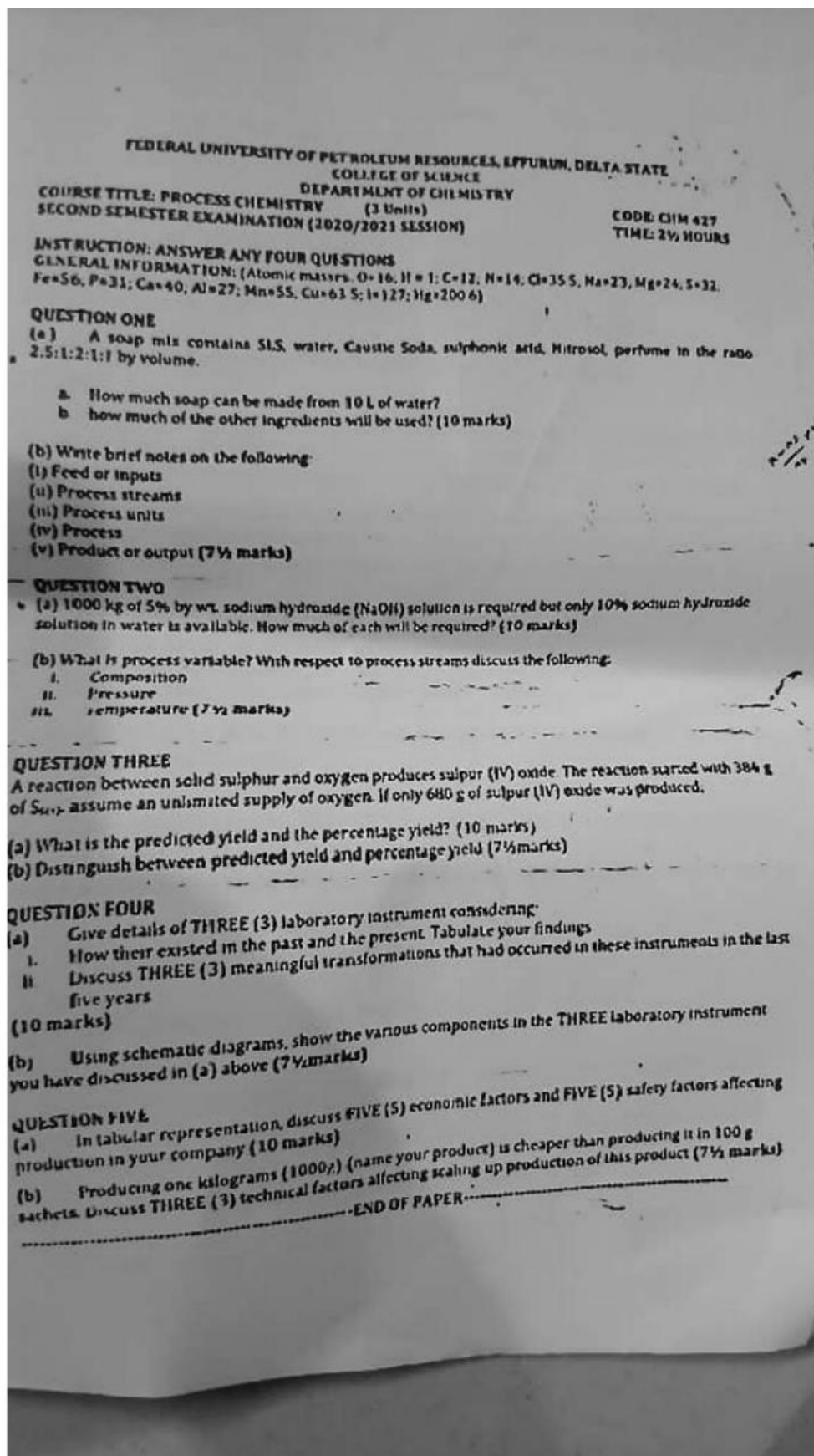
¢ flow and composition of the product.

END OF QUESTIONS

Paper 2 of 4 — CHM 427 PQ

Level 400 • Second Semester • PastQuestion

This source paper is a scanned document; its pages are reproduced below.



QUESTION FOUR

- (a) Discuss the fate of polychlorinated biphenyls (PCBs) in the environment; list any five of their applications and five health effects. 10marks
- (b) What are polycyclic aromatic hydrocarbons (PAHs) and how are they formed? 7marks
- (c) Discuss the sources of PAHs in the environment and list four of their properties. 8marks

25MARKS**QUESTION FIVE**

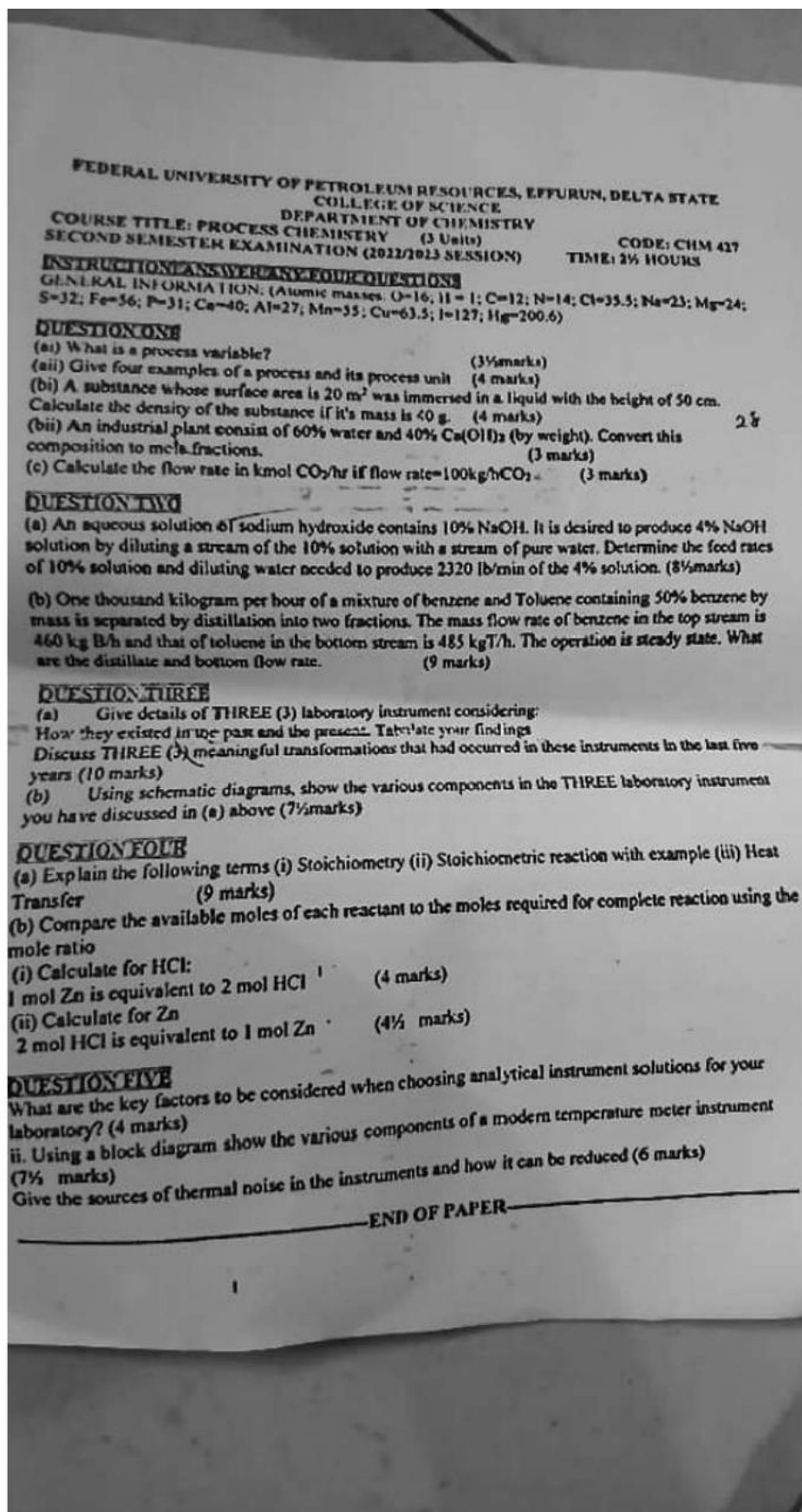
- (a) The potential of undiscovered oil/ gas in Nigeria is highly dependent on a broad spectrum of factors. Explain three of these factors and how they could be harnessed for the benefit of the citizens of Nigeria. 10marks
- (b) Briefly explain the term, 'Memorandum of understanding' as applied in the oil industry 10marks.
- (c) What is a condensate? 5marks

25MARKS

Paper 3 of 4 — CHM 427 PQ 2023

Level 400 • Second Semester • PastQuestion

This source paper is a scanned document; its pages are reproduced below.



Paper 4 of 4 — CHM 427 PastQuestion 2018/2019

Level 400 • Second Semester • PastQuestion

This source paper is a scanned document; its pages are reproduced below.

